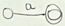
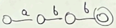
L:

T1: a :

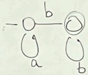


T2: a b b :

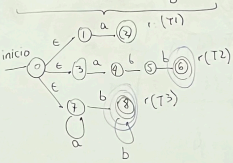


T3:

a* b+



T1 y T2/T3



Destados: { A✓, B✓, C✓, D✓, E✓, Error✓, F✓ }

C-ε(0) = {0, 1, 3, 7} A

P/A: a
U: C-ε(m(A, a)) = C-ε({2, 4, 7}) = {2, 4, 7} B
b
U: C-ε(m(A, b)) = C-ε({8}) = {8} C

P/B: a
C-ε(m(B, a)) = {7} D
b
C-ε(m(B, b)) = {5, 8} E

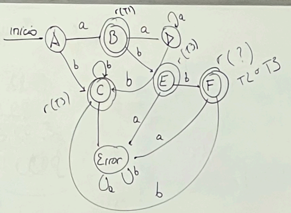
P/C:
a
C-ε(m(C, a)) = {–} Error
b
C-ε(m(C, b)) = {8} C

P/D:
a
C-ε(m(D, a)) = {7} D
b
C-ε(m(D, b)) = {8} C

P/E:
a
C-ε(m(E, a)) = {–} Error
b
C-ε(m(E, b)) = {6, 8} F

P/F:
a
C-ε(m(F, a)) = {–} Error
b
C-ε(m(F, b)) = {8} C

Trans	a	b
A	B	C
B	D	E
C	Error	C
D	D	C
E	Error	F
F	Error	C
Error	Error	Error

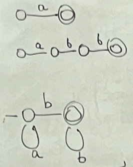


L:

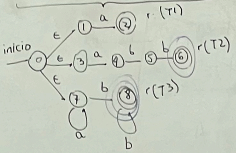
T1: a

T2: a b b

T3: a* b+



T1 y T2/T3



Destados: {A✓, B✓, C✓, D✓, E✓, Error✓, F✓}

C-ε(0) = {0,1,3,7} A

P/A: a
U: C-ε(m(A,a)) = C-ε({2,4,7}) = {2,4,7} B
b
U: C-ε(m(A,b)) = C-ε({8}) = {8} C

P/B: a
C-ε(m(B,a)) = {7} D
b
C-ε(m(B,b)) = {5,8} E

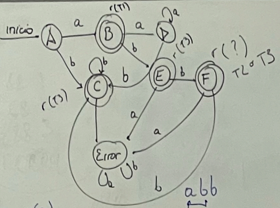
P/C:
a
C-ε(m(C,a)) = {–} Error
b
C-ε(m(C,b)) = {8} C

P/D:
a
C-ε(m(D,a)) = {7} D
b
C-ε(m(D,b)) = {8} C

P/E:
a
C-ε(m(E,a)) = {–} Error
b
C-ε(m(E,b)) = {6,8} F

P/F:
a
C-ε(m(F,a)) = {–} Error
b
C-ε(m(F,b)) = {8} C

Trans	a	b
A	B	C
B	D	E
C	Error	C
D	D	C
E	Error	F
F	Error	C
Error	Error	Error



```
S = C-ε(S0)
c = sgleInput()
while(c != eof) {
    S = C-ε(m(S,c));
    c = sgleInput();
}
if(S∩F != 0) return(OK)
return(FALSE)
```

T1/47/71

abb

S: {0,1,3,7}

S: {2,4,7}

S: {5,8}

S: {6,8}

Minimización de estados de un AFD

1.- Definir un conj. Π formado por 2 grupos de estados. Π = {S, F}

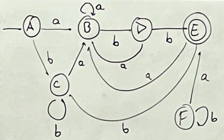
2.- Mientras se obtenga una Π nueva

Para cada estados S y T de c/grupo en Π: S y t se separan si para a, S y T van a estados sgtes que estén en grupos diferentes

3.- El estado inicial es el grupo donde esté el est. inicial original

- Estados finales son los grupos de est. finales

4.- Eliminar posibles estados inalcanzables. ✓



$\Pi = \{F, S\} = \{\underset{I}{\{E\}}, \underset{II}{\{A, B, C, D\}}\}$

$\Pi_{nueva} = \{\underset{I}{\{E\}}, \underset{II}{\{A, B, C\}}, \underset{III}{\{D\}}\}$

$\Pi_{nueva} = \{\underset{I}{\{E\}}, \underset{II}{\{A, C\}}, \underset{III}{\{D\}}, \underset{IV}{\{B\}}\}$

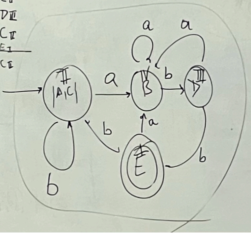
$\Pi_{nueva} = \{\{E\}, \{A, C\}, \{D\}, \{B\}\}$

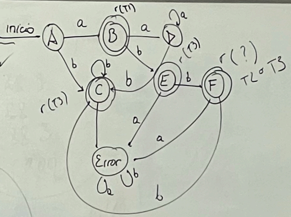
	a	b
A	B II	C I
B	B II	D II
C	B II	C II
D	B II	E I
E	B II	C I

	a	b
A	B III	C II
B	B III	D III
C	B III	C II
D	D II	E I
E	D II	C II

	a	b
→ A	B IV	C II
B	B IV	D III
→ C	B IV	C II
D	B IV	E I
E	B IV	C II

	a	b
A		
C	✓	





Minimización de estados de un AFD

1: Definir un conj. Π formado por 2 grupos de estados. $\Pi = \{S, F\}$

2.- Mientras se obtenga una Π nueva
Para cada estados s y T de c/grupo en Π: s y t se separan si para a, s y T van a estados sgtes que estén en grupos diferentes

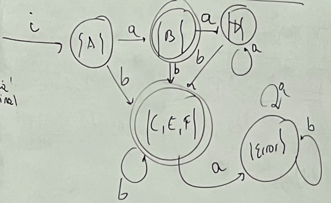
3.- El estado inicial es el grupo donde esté el est. inicial original
- Estados finales son los grupos de est. finales

4.- Eliminar posibles estados inalcanzables. ✓

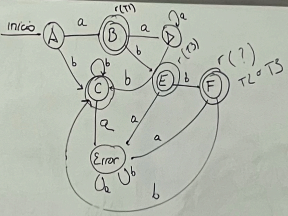
$$\Pi_c \{ \{B, C, E, \underset{I}{F}\}, \{A, D, \underset{II}{Error}\} \}$$

$$\Pi_n = \{ \{D, C, E, \underset{I}{F}\}, \{\underset{II}{A}\} \{\underset{III}{D}\} \{\underset{IV}{Error}\} \}$$

$$\Pi_n = \{ \{\underset{I}{B}\} \{C, E, \underset{V}{F}\}, \{\underset{II}{A}\}, \{\underset{III}{D}\}, \{\underset{IV}{Error}\} \}$$



	a	b
~~A~~	~~B~~	~~C~~
~~B~~	~~D$_{III}$~~	~~E~~
C	Error$_{IV}$	C
~~D~~	~~D$_{III}$~~	~~C~~
E	Error$_{IV}$	F
F	Error$_{IV}$	C
~~Error~~	~~Error$_{II}$~~	~~Error$_{IV}$~~



P/cada estado de un AFN ⟷ AFD

↓

BNF

1. Cada estado i, se representa con A_i

2.- Si ∃ i — j con la entrada a.

$A_i \to a A_j$

Si ∃ i — j con ε

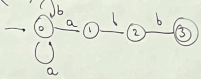
$A_i - A_j$

3- Si i es un estado final.

$A_i \to \epsilon$

4- Si i es inicial, A_i es inicial tmb.

$(a|b)^* abb.$



} expr reg

AFN/AFD

BNF eq.

$A_0 \to a A_0 \mid a A_1 \mid b A_0$

$A_1 \to b A_2$

$A_2 \to b A_3$

$A_3 \to \epsilon$

Sn

FC ≡ Inde...

F.C.I ≡ [illegible]

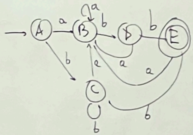
BNF + A.S

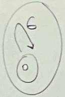
o

Der.

$A_i - A_j \mid a A_x$

$A_j - a$





|G,δ|

$A_A - a A_B \mid b A_C$

$A_B \to a A_B \mid b A_D$

$A_C - a A_B \mid b A_C$

$A_D - a A_B \mid b A_E$

$A_E - a A_B \mid b A_C \mid \epsilon$

R.I ✓ X

F.C.D ✓ X

F.C.I. V

P/cada estado de un AFN ✓ ⟷ AFD ✓ → BNF ✓

1. Cada estado i, se representa con A_i

2.- Si ∃ i – j con la entrada a.

$A_i \to a A_j$

Si ∃ i – j con ε

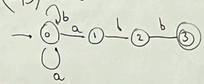
$A_i \to A_j$

3.- Si i es un estado final.

$A_i \to \epsilon$

4 – Si i es inicial, A_i es inicial tmb.

$(a|b)^* abb.$



BNF eq.

$A_0 \to a A_0 | a A_1 | b A_0$

$A_1 \to b A_2$

$A_2 \to b A_3$

$A_3 \to \epsilon$

3+4-2

$(dig + dig | dig - dig)^*$

[0-9]

$A_i \to A_j | a A_X$

$A_j \to a$

Expr reg

AFN/AFD

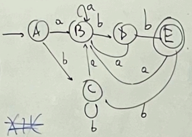
Sn

FC ≡ Inde...

F.C.I. ≡ (no)

BNF + A.S o Der.

a^*



~~AHC~~

V|V

$A_A \to a A_B | b A_C$

$A_B \to a A_B | b A_D$

$A_C \to a A_B | b A_C$

$A_D \to a A_B | b A_E$

$A_E \to a A_B | b A_C | \epsilon$

|6,δ|

R.I. ✓ X

F.C.D ✓ X

F.C.I. V